

GIT BRANCH

Definition

A Git branch is a separate line of development and workspace in a repository used to work on changes independently without affecting the main code.

Uses of Git Branch

- Create separate workspace for development
- Develop new features separately
- Fix bugs safely
- Test code without affecting main branch
- Allow multiple developers to work simultaneously
- Merge changes after successful testing

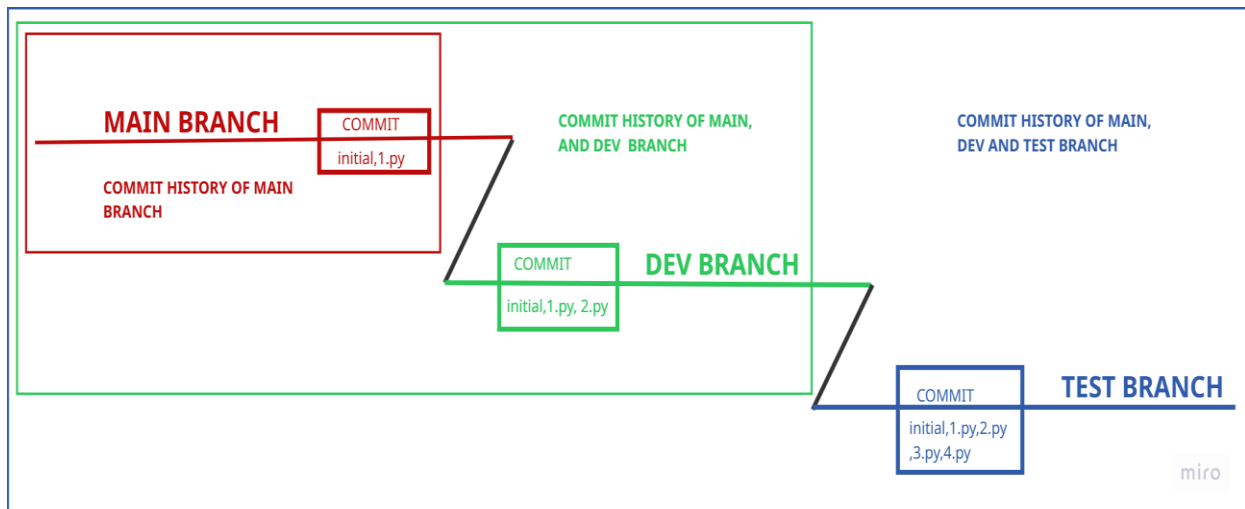
Features of Git Branch

- Lightweight and fast to create
- Multiple branches in one repository
- Independent development environment
- Easy switching between branches
- Supports merging changes into main branch
- Helps in organized and parallel development

Git Branch Commands with Definition

Command	Definition
git branch	Show all branches in the repository
git branch branch-name	Create a new branch
git checkout branch-name	Switch to another branch
git checkout -b branch-name	Create and switch to a new branch
git merge branch-name	Merge a branch into the current branch
git branch -d branch-name	Delete a branch

GIT BRANCH ARCHITECTURE



HANDS ON FOR GIT BRANCH

Screenshot: Create New Repository

- Click New repository
- Enter repository name
- Enable Add README file
- Click Create repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * Repository name *

repo-branch is available.

Great repository names are short and memorable. How about [refactored-giggle?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility * Choose who can see and commit to this repository

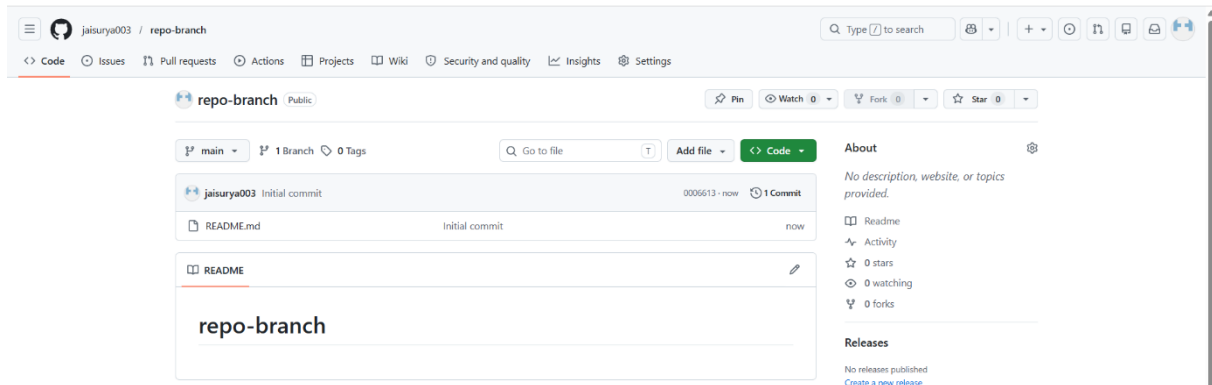
Add README READMEs can be used as longer descriptions. [About READMEs](#) On ☒

Add .gitignore .gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

Add license Licenses explain how others can use your code. [About licenses](#) No license

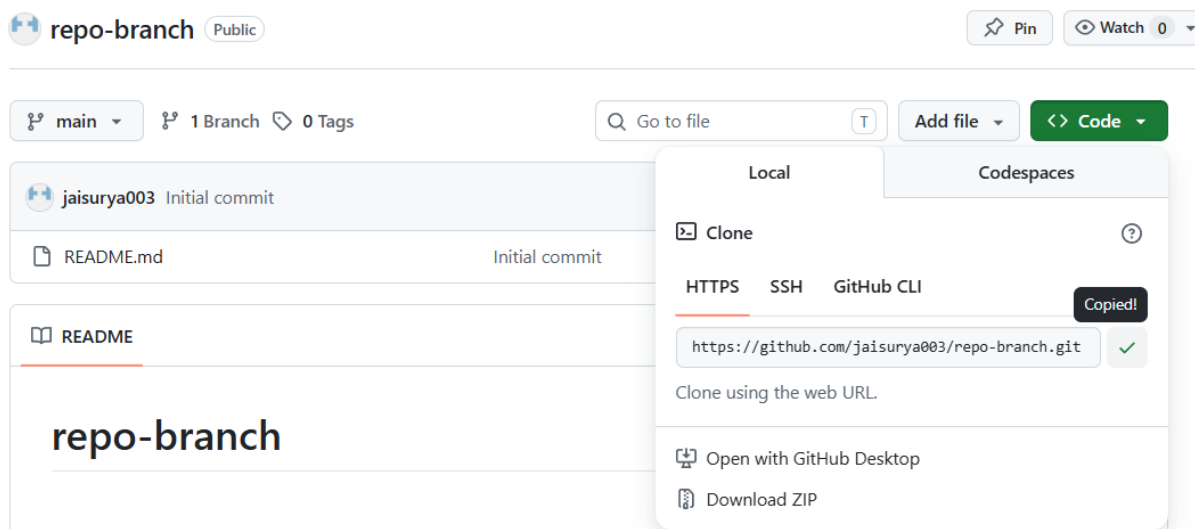
Screenshot: Repository Created

- Repository created successfully
- Repository dashboard opened



Screenshot: Copy Repository Link

- Click Code
- Copy HTTPS repository link



Screenshot: Clone Repository in Git Bash

- Open Git Bash
- Execute `cd Downloads`
- Execute `git clone <repository-link>`
- Execute `cd <repository-name>`

MINGW64:/c/Users/JAISURYA/downloads/newfolder/repo-branch

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~  
$ cd downloads  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads  
$ mkdir newfolder  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads  
$ cd newfolder  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder  
$ git clone https://github.com/jaisurya003/repo-branch.git  
Cloning into 'repo-branch'...  
remote: Enumerating objects: 3, done.  
remote: Counting objects: 100% (3/3), done.  
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (3/3), done.  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder  
$ cd repo-branch
```

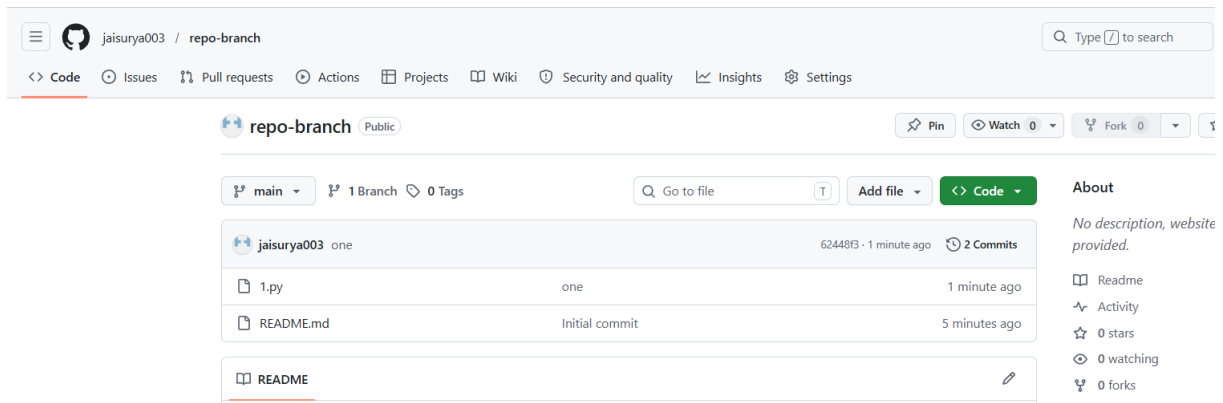
Screenshot: Create and Push File

- Execute touch 1.py
- Execute git add 1.py
- Execute git commit -m "one"
- Execute git push origin main

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)  
$ touch 1.py  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)  
$ git add 1.py  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)  
$ git commit -m "one"  
[main 62448f3] one  
1 file changed, 0 insertions(+), 0 deletions(-)  
create mode 100644 1.py  
  
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)  
$ git push origin main  
Enumerating objects: 4, done.  
Counting objects: 100% (4/4), done.  
Delta compression using up to 12 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 266 bytes | 133.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To https://github.com/jaisurya003/repo-branch.git  
0006613..62448f3 main -> main
```

Screenshot: File Added in Main Branch

- Refresh GitHub repository
- Verify 1.py file is available in main branch



Screenshot: Create Git Branch

- Execute git branch
- Execute git branch dev

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch
* main

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch dev
```

Screenshot: Push Changes to Dev Branch

- Execute git branch
- Execute git checkout dev
- Execute touch 2.py
- Execute git add 2.py
- Execute git commit -m "two"
- Execute git push origin dev

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch
dev
* main

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git checkout dev
Switched to branch 'dev'

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ touch 2.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git add 2.py

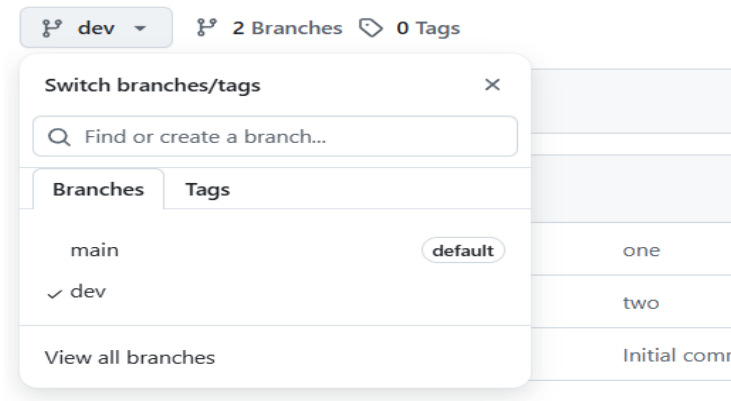
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git commit -m "two"
[dev 71214bd] two
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 2.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git push origin dev
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 261 bytes | 261.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/jaisurya003/repo-branch/pull/new/dev
remote:
To https://github.com/jaisurya003/repo-branch.git
 * [new branch]      dev -> dev

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$
```

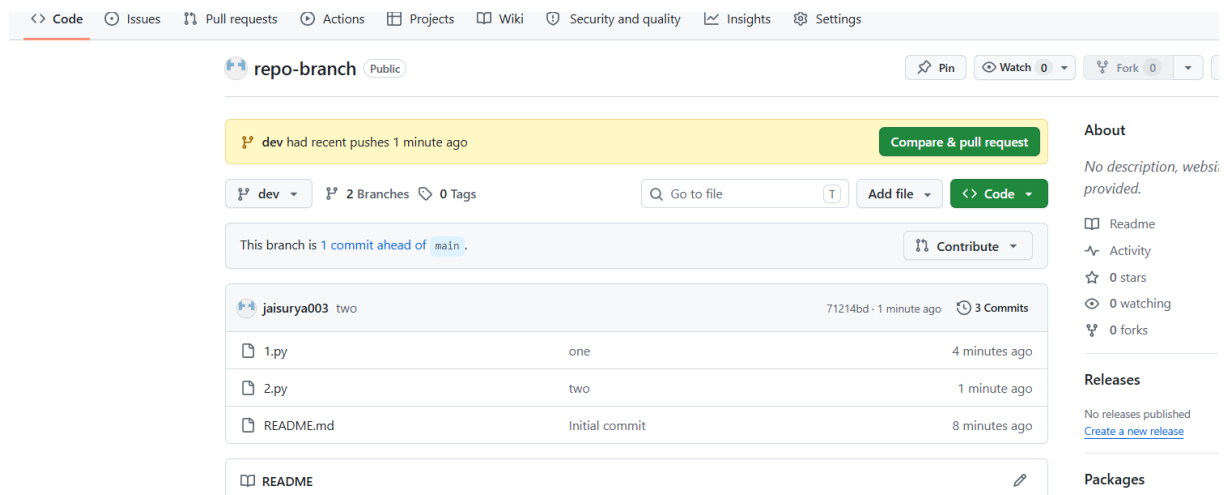
Screenshot: Dev Branch Created in GitHub

- Go to GitHub repository
- Switch branch to dev



Screenshot: Dev Branch Files Verified

- Switch to dev branch on GitHub
- Confirm 1.py, 2.py, and README.md are available in dev branch
- Files successfully present in dev branch



Screenshot: Create and Push Test Branch

- Execute git checkout -b test
- Create 3.py and 4.py
- Execute git add .
- Execute git commit -m "test branch files"
- Execute git push origin test

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git checkout -b test
Switched to a new branch 'test'

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ touch 3.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ touch 4.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git add .
```

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git commit -m "four and five"
[test 2ae7cd3] four and five
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 3.py
create mode 100644 4.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git push origin test
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 274 bytes | 274.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test' on GitHub by visiting:
remote:   https://github.com/jaisurya003/repo-branch/pull/new/test
remote:
To https://github.com/jaisurya003/repo-branch.git
 * [new branch]   test -> test

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$
```

Screenshot: Test Branch Created in GitHub

- Go to GitHub repository
- Switch to test branch

The screenshot shows the GitHub interface for a repository named 'repo-branch'. The 'test' branch is selected, and a dropdown menu for switching branches is open, showing 'test' as the current branch. The commit history table is visible below.

Commit Hash	Commit Message	Time Ago
2ae7cd3	four and five	1 minute ago
one		7 minutes ago
two		4 minutes ago
four and five		1 minute ago
four and five		1 minute ago
Initial commit		11 minutes ago

Screenshot: Test Branch Files Verified

- Switch to test branch on GitHub
- Confirm 1.py, 2.py, 3.py, 4.py, and README.md are available
- All files successfully present in test branch

The screenshot shows the GitHub interface for a repository named 'repo-branch'. The 'test' branch is selected, and the file list shows the following files and their commit history:

File	Commit	Time
1.py	one	7 minutes ago
2.py	two	4 minutes ago
3.py	four and five	1 minute ago
4.py	four and five	1 minute ago
README.md	Initial commit	11 minutes ago

The 'test' branch is 2 commits ahead of 'main'. The repository has 3 branches and 0 tags. The 'test' branch has recent pushes 53 seconds ago. The repository is public and has 0 stars, 0 forks, and 0 watchers.